# 25CSSE01C

# Software Engineering I

Phase 1 - Report

**Project Title:** Pharmacy Delivery System

**Group #:** 40

**T. A. Name:** Zeina

# Table of Contents

1. Contribution Table: ..... 2
2. Project Description: ..... 3
3. Functional Requirements: ..... 4
4. Non-Functional Requirements: ..... 5
5. Context Diagram (Group): ..... 5
6. Use Case Diagram (Group): ..... 6
7. Use Case Scenario (Individual): ..... 7
   7.1. Manage Inventory (Youssef Adel – 258270 ) ..... 8
   7.2. Assign Courier (Tarek Saeed - 252382) ..... 8
   7.3. Manage Profile (Yosef Osama - 255796) ..... 9
   7.4. Track Order Statues (Yousif Hafez - 258612) ..... 11
   7.5. Manage Order (Yousef Mohiey - 248679) ..... 12

# 1. Contribution Table:

| ID | Student Name | Contribution |
|---|---|---|
| **258270** | Youssef Adel | Functional Requirements (FR-01), Use Case Scenario 1 |
| **248679** | Yousef Mohamed Mohiey | Functional Requirements (FR-02), Use Case Scenario 5 |
| **258612** | Yousif Hafez Momtaz | Functional Requirements (FR-04), Use Case Scenario 4 |
| **255796** | Yosef Osama Gad | Functional Requirements (FR-05), Use Case Scenario 3 |
| **252382** | Tarek Saeed | Functional Requirements (FR-03), Use Case Scenario 2 |

## 2. Project Description:

Pharmacy Delivery System

The system must maintain an accurate inventory of medicines, including pricing and stock levels, alongside a secure database of customer profiles and verified delivery addresses. Students must define a multi-stage

OrderWorkflowthatbegins with orderplacementand continues through to "Dispatch" and "FinalDelivery." A core requirement is the Courier Assignment Module, which allows administrators to assign delivery personnel to specific orders based on priority or location. Additionally, the system must provide Real-time Status Tracking (e.g., Pending, Out for Delivery, Delivered) to ensure order accuracy and provide transparency for both the pharmacy and the end-user.

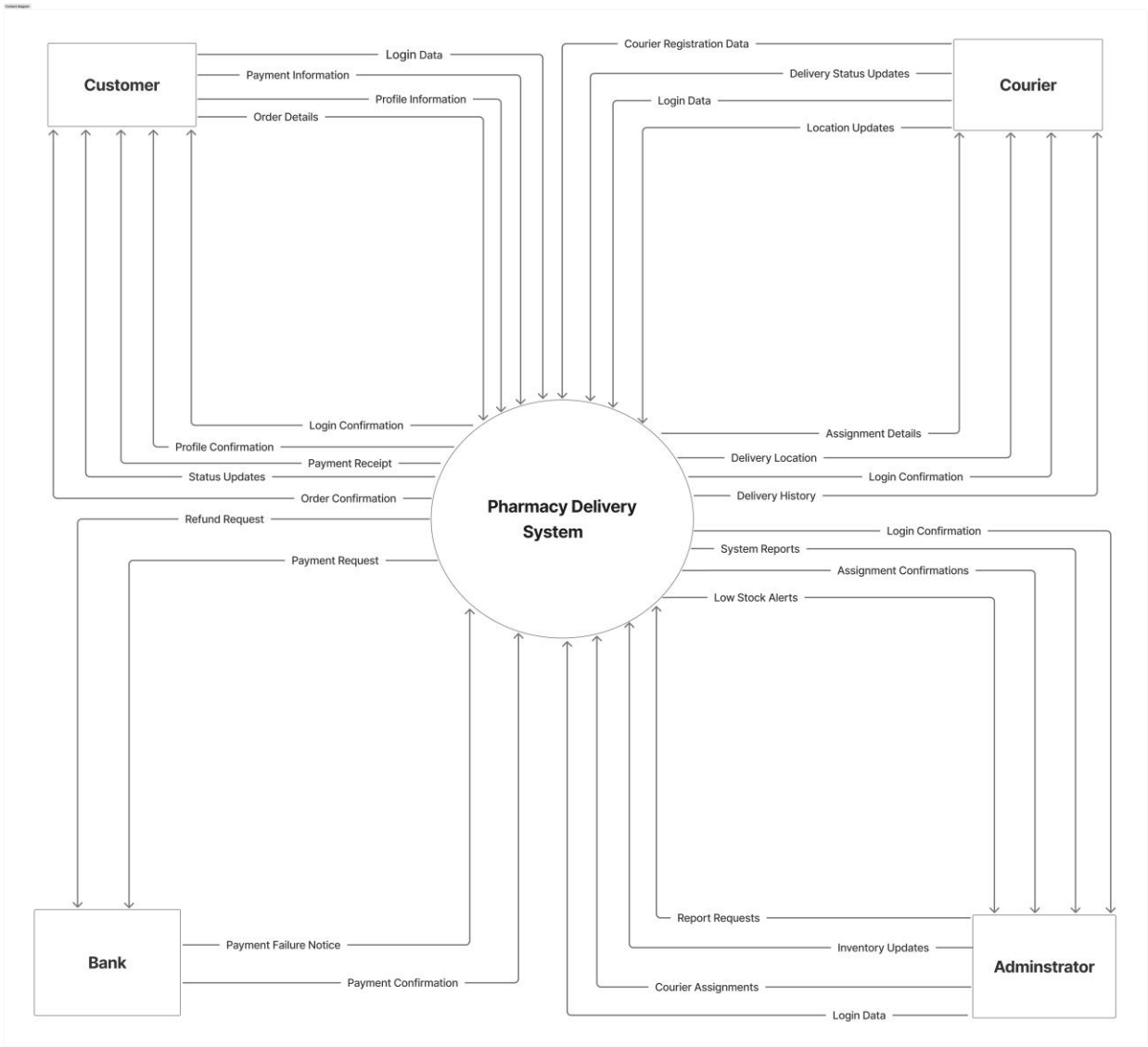
## 3. Functional Requirements:

| Function | Description |
|---|---|
| **Manage Inventory** | The system shall allow authorized pharmacy staff to manage medicine inventory including pricing, stock levels, and product details. |
| **Manage Order** | The system shall allow customers to process orders from placement through to dispatch and final delivery. |
| **Assign Courier** | The system shall allow administrators to assign couriers to order based on priority or location. |
| **Track Order-Status** | The system shall allow customers and staff to track order status with real-time updates (Pending, Out for Delivery, Delivered). |
| **Manage Profile** | The system shall allow customers to manage customer profiles, including verified delivery addresses and contact information. |

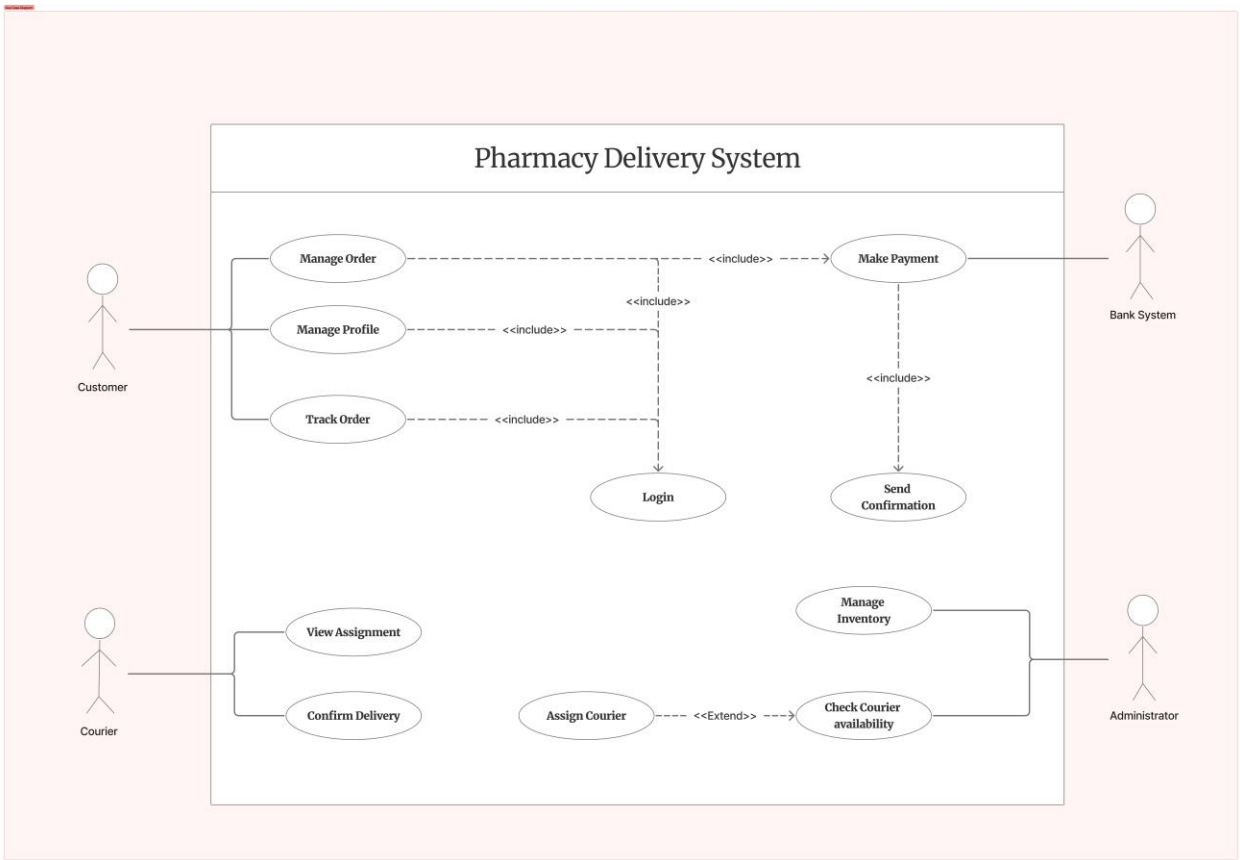
## 4. Non-Functional Requirements:

1. Performance**:** Order status updates must reflect within **5 seconds** of status change.
2. **Security:** Customer data must be **encrypted using AES-256 standards**.
3. **Availability:** System must have **99.9% availability** during peak hours (6 PM - 10 PM).
4. **Compliance:** System must comply with **local pharmacy data protection laws**.
5. **Usability:** First-time users shall complete an order in **under 5 minutes** without assistance.

# 5. Context Diagram (Group):



# 6. Use Case Diagram (Group):



# 7. Use Case Scenario (Individual):

## 7.1. Manage Inventory (Youssef Adel – 258270 )

| | |
|---|---|
| **ID:** | UC_01 |
| **Title:** | Manage Inventory |
| **Description:** | Admin manages medicine inventory including stock and pricing. |
| **Priority** | High |
| **Primary Actor:** | Admin/Pharmacist |
| **Secondary Actor:** | N/A |
| **Preconditions:** | • Admin is logged in<br>• Admin has inventory management permissions |
| **Postconditions:** | • Inventory database is updated<br>• Stock levels reflect changes<br>• Confirmation message displayed to Admin |
| **Main Success Scenario:** | **1.** Admin navigates to inventory management page<br>**2.** Admin searches for medicine by name or ID<br>**3.** Admin views current stock<br>**4.** Admin enters new quantity/price and clicks (Save Changes)<br>**5.** System saves changes and displays confirmation |
| **Extensions:** | • 2a - Medicine not found ( "Add new medicine" option displayed )<br>• 3a - Invalid data entered ( Error message displayed )<br>• 5a - Database error ( Error message, changes not saved )<br>• 6a - Admin clicks 'Cancel' ( Changes discarded, returns to inventory page ) |

## 7.2. Assign Courier (Tarek Saeed - 252382)

| | |
|---|---|
| **ID:** | UC_02 |
| **Title:** | **Assign Courier** |
| **Description:** | The system allows administrators to assign delivery personnel to specific orders. |
| **Priority** | High |
| **Primary Actor:** | Administrator |
| **Secondary Actor:** | Courier |
| **Pre-conditions:** | Order status is "Pending".<br><br>Couriers are available. |
| **Post-conditions:** | Order status updates to "Dispatch".<br><br>Courier receives assignment notification. |
| **Main Success Scenario:** | Admin views pending orders list.<br><br>Admin selects order based on priority/location.<br><br>Admin selects available courier.<br><br>System assigns courier to order.<br><br>System notifies courier. |
| **Extensions:** | No couriers available. System alerts Admin to wait or outsource.<br><br>Courier is offline. System suggests next available courier.<br><br>Assignment conflict (Courier already assigned). System blocks action. |

## 7.3. Manage Profile (Yosef Osama - 255796)

| | |
|---|---|
| **ID:** | UC_03 |
| **Title:** | Manage profile |
| **Description:** | Allows customers to view, update, and maintain their personal information |
| **Priority** | High |
| **Primary Actor:** | Customer |
| **Secondary Actor:** | Bank System |
| **Pre-conditions:** | The Customer needs to have a system registration.<br>The Customer needs to log in successfully (Register/Login). |
| **Post-conditions:** | The system updates and saves the customer profile data.<br>The system keeps track of profile modifications. |
| **Main Success Scenario** | The Customer opens the "Manage Profile" section<br>The system shows the most recent profile data.<br>The customer chooses which information (personal information, addresses, payment methods, or preferences) to update.<br>The system displays an editable form<br>The Customer adds or modifies data<br>The system verifies the data entered.<br>The system stores updated data<br>The system verifies that the update was successful.<br>The Customer sees the confirmation message |
| **Extensions:** | Incorrect information entered:The system shows an error message<br>Information is corrected by the customer<br><br>Error in the system while saving .The system shows an error message. The client may cancel or try again. Resuming the end use case or step five |

## 7.4. Track Order Statues (Yousif Hafez - 258612)

| | |
|---|---|
| **ID:** | UC_04 |
| **Title:** | Track Order Statues |
| **Description:** | The system shall display the current order status in real time, showing stages such as Pending, Processing, Out for Delivery, and Delivered. |
| **Priority** | High |
| **Primary Actor:** | Customer |
| **Secondary Actor:** | Admin / Pharmacy Staff / Courier |
| **Pre-conditions:** | • User should have already placed an order.<br>• Order should exist in the system.<br>• User should be logged into the system or have access to the order details. |
| **Post-conditions:** | • The system displays the latest order status successfully.<br>• User is informed of the current delivery stage of the order. |
| **Main Success Scenario:** | **1. User opens the system/application.**<br>**2. User navigates to the order history or current orders page.**<br>**3. User selects the order they want to track.**<br>**4. System retrieves the latest order information.**<br>**5. System displays the current order status (Pending, Processing, Out for Delivery, Delivered).**<br>**6. User views the order progress** |
| **Extensions:** | **1. The selected order may be invalid or unavailable**, so the system displays an error message and asks the user to choose a valid order.<br><br>**2. The system may fail to retrieve the latest order status**, so an error message is displayed asking the user to try again later.<br><br>**3. Network or connection problems may occur while loading the order status**, so the system asks the user to check the connection and retry. |

## 7.5. Manage Order (Yousef Mohiey - 248679)

| | |
|---|---|
| **ID:** | UC_05 |
| **Title:** | Manage Order |
| **Description:** | This function allows customers to place new orders, modify existing orders, and cancel orders (if status is Pending). |
| **Priority** | High |
| **Primary Actor:** | Customer |
| **Secondary Actor:** | Bank System |
| **Pre-conditions:** | • Customer is registered and logged in<br>• Medicines are in stock (for new orders)<br>• Order exists in system (for modify/cancel) |
| **Post-conditions:** | • **For Place:** Order created with status "Pending", Payment confirmed<br>• **For Modify:** Order details updated, notification sent<br>• **For Cancel:** Order status changed to "Cancelled" |
| **Main Success Scenario:** | 1. Customer logs in and opens Manage Orders.<br>2. System shows three options: Place, Modify, or Cancel.<br>3. Customer picks one option:<br>a. Place: Customer adds medicines and delivery address.<br>b. Modify: Customer selects a Pending order and changes details.<br>c. Cancel: Customer selects a Pending order and confirms cancel.<br>4. System checks stock and order status.<br>5. System processes payment through Bank System.<br>6. System updates the order status.<br>7. System sends confirmation to customer. |

| | |
|---|---|
| **Extensions:** | 3a. Stock not available. System alerts customer. Customer changes order.<br>3b. Order already shipped. System blocks cancel/modify. Shows error.<br>5a. Payment fails. System asks retry. Customer selects new payment.<br>5b. Bank System unavailable. System saves as Pending Payment. Notifies customer to retry later. |